

ensures transparency to the client.

- **Timely delivery:** Usually, the duration of a sprint is fixed and varies between one and four weeks, which forces the team to deliver features rapidly and frequently. This also helps product owners to predict the costs involved in the development and keep these under check.
- **Changing requirements:** The Agile methodology also allows teams to incorporate changes in the requirements at an early stage of the development cycle, which helps companies to develop high end products without overshooting their budgets.
- **User focused:** Instead of test cases, the Agile model employs user stories that have business and user focused acceptance criteria. This helps teams to understand the needs of the users and deliver products that can be beta tested in time, so that the necessary changes can be done at the earliest.

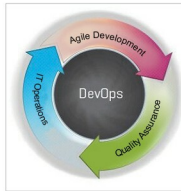


Figure 1: Agile and DevOps complement each other with the support of the QA and IT operations teams



Figure 2: Wall of confusion between the development and operations teams

Steps in the Agile approach

Let's look at the steps involved in implementing the Agile methodology.

1. **Discovery:** To develop a high quality product, one needs to have a clear vision and considerable experience in the technology used in that project. Discovery sessions are significant, since they are the basis for all the upcoming activities in the sprint. During these sessions, the clients' goals, the users' expectations and the business challenges are understood deeply so that no ambiguity remains in the minds of the team, regarding the product.
2. **Product backlog:** The result of successful discovery sessions is product backlog, which contains a list of all the features that need to be developed. These features are then classified on the basis of priority by the product owner (in discussion with the client), so that high priority features can be developed, tested and delivered first.
3. **Iterations:** After the high-level product backlog is finalised along with the priority, sprints are planned and work begins on the features mentioned in the backlog.



Note: Every successive sprint in Agile is both iterative and incremental. It is iterative in the sense that it provides improvements based on the experience gained in the previous sprints, and incremental because it adds new features to the system.

4. **Cycle:** If all the features are completed and tested successfully, then the cycle stops; otherwise, additional

sprints are planned to carry out the remaining work.

Agile and DevOps: The connection

Agile and DevOps – these two terms have become the buzzword these days. Though these two words are used interchangeably, there's a stark difference between them. Agile is mainly concerned with software development and the processes or steps involved in it, whereas DevOps comes into the picture after a high quality product has been developed, i.e., it is about the deployment and management of software. The term DevOps is derived from two words – development and operations. Before delving deeper into the details of DevOps, let's see how it emerged in the IT scene.

We have seen how organisations reaped benefits by implementing the Agile methodology, but this model also had some hitches, which are listed below:

- There were chances of incompatibility between old features and new features during integration.
- Often, budget goals and deadlines were missed.
- There was a lack of cooperation between the development and IT operations teams.

Usually, whenever any product is released or any service is made live by an IT organisation, two departments come together to support this release – the development and operations teams. Yet, there is a lack of coordination between development activity and operations activity. The development team feels it is being paid to bring about 'change', whereas the operations team is looking at stability and considers 'change' its enemy. This conflict in mindsets often leads to inefficiency and overhead costs for the company. DevOps is a practice employed to smoothen the IT service delivery by promoting communication between development and operations teams, which is essential to increase a company's productivity. It helps the company to continually deliver software with highly stable features, faster and more frequently.

DevOps brings more flexibility to the Agile methodology and leverages its productivity. It widens the scope of Agile principles by including operations teams in its ambit instead of stopping the Agile cycle at code check-in only. So, you can deduce that Agile principles and processes can be employed as a part of DevOps. In layman's language, we can say that by using the Agile methodology, high-end products are developed and by implementing DevOps, the developed products are deployed in a timely manner. So the Agile model and DevOps complement each other, but are totally different from one another.